34. The Desert Ant

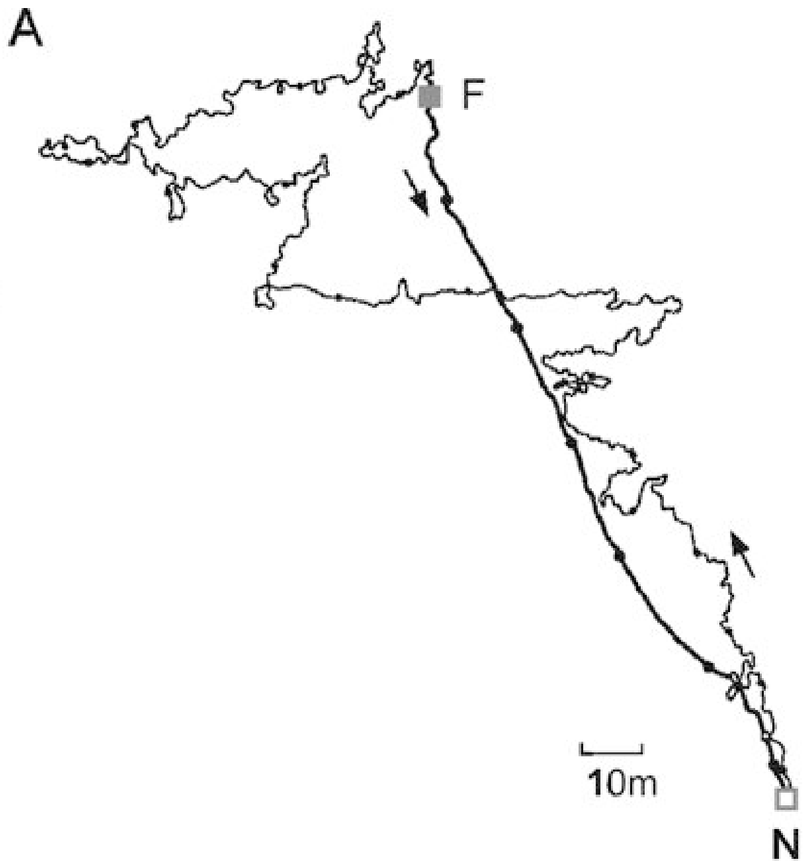
The Ant



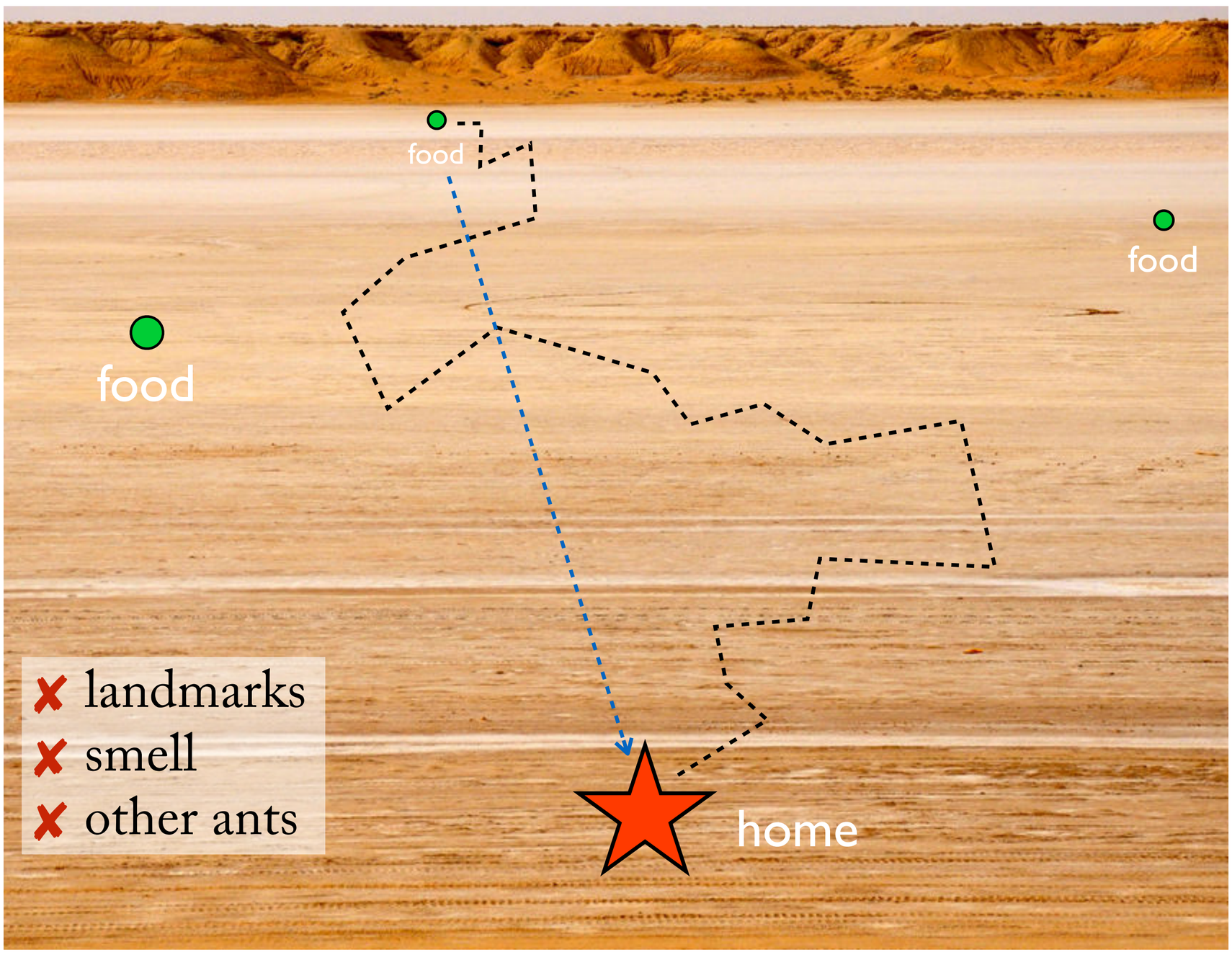


250,000 neurons

The Journey

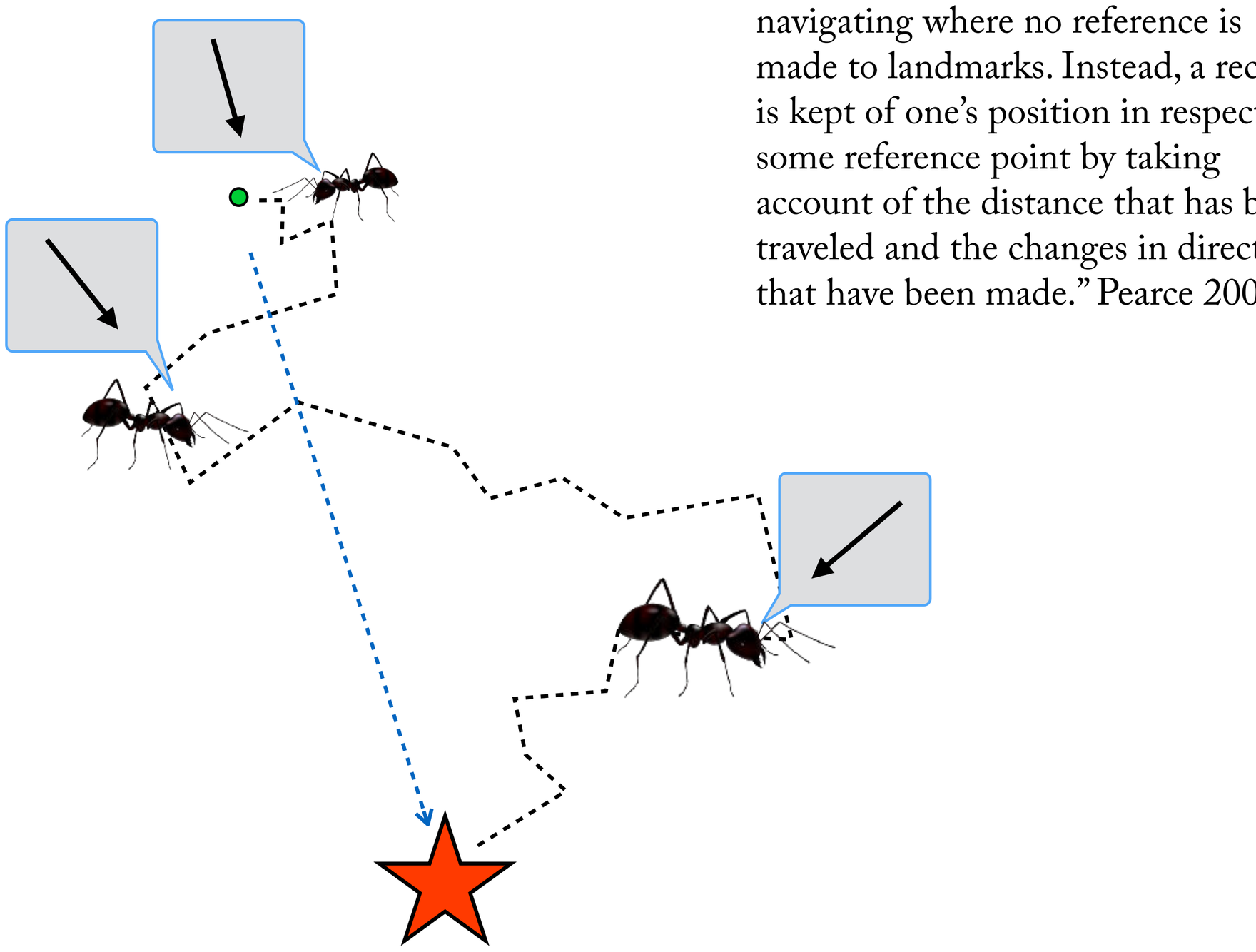


"Scientists observed that instead of retracing their winding route back home, they head straight back to the nest. How does the ant keep track of its current location relative to home? Path integration is the name given to the process thought to be used by animals (human and non-human alike) for what some might call dead reckoning." — SA



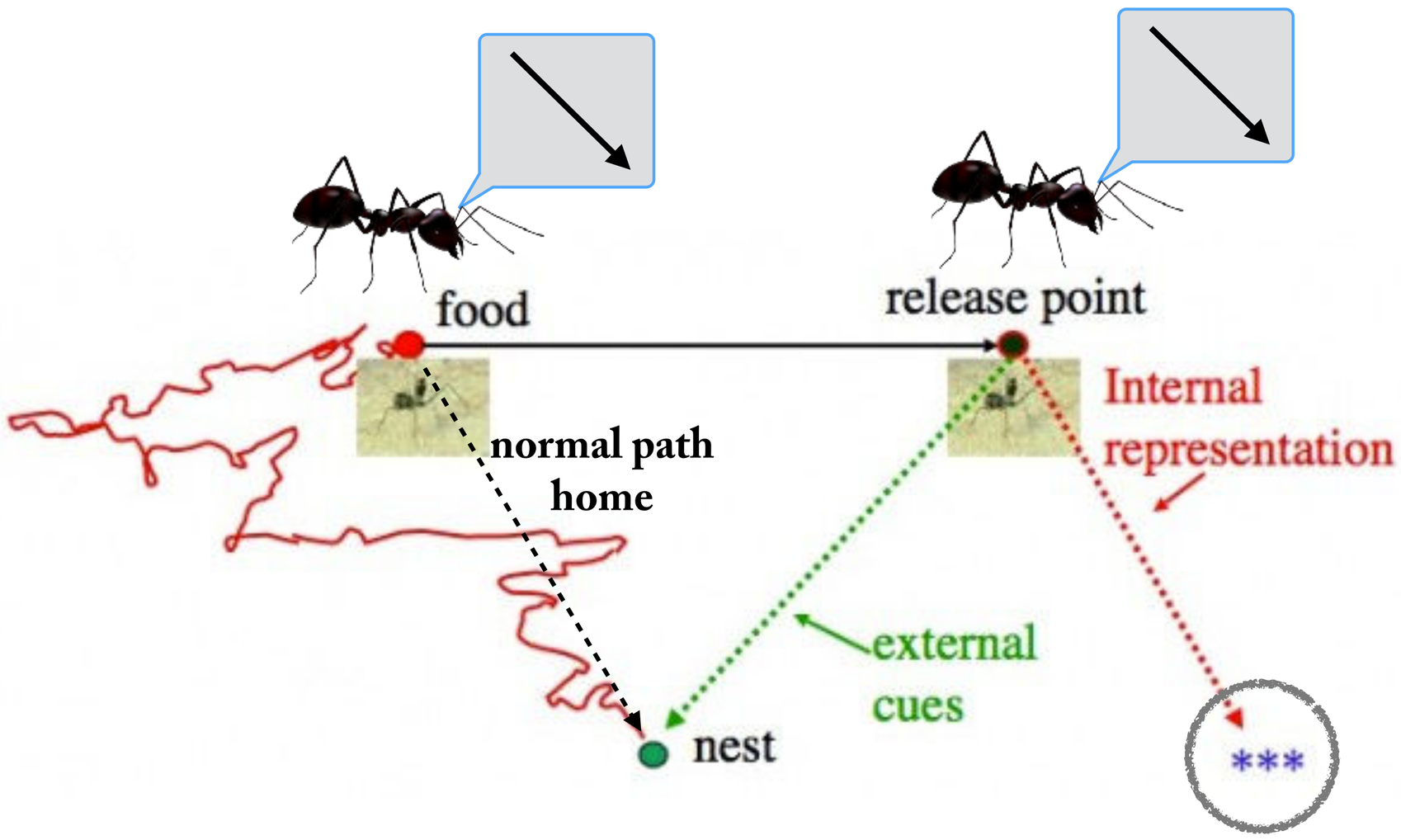
Ant = ~ 0.5 cm
Travel = >150 m
= over 20,000x
their length

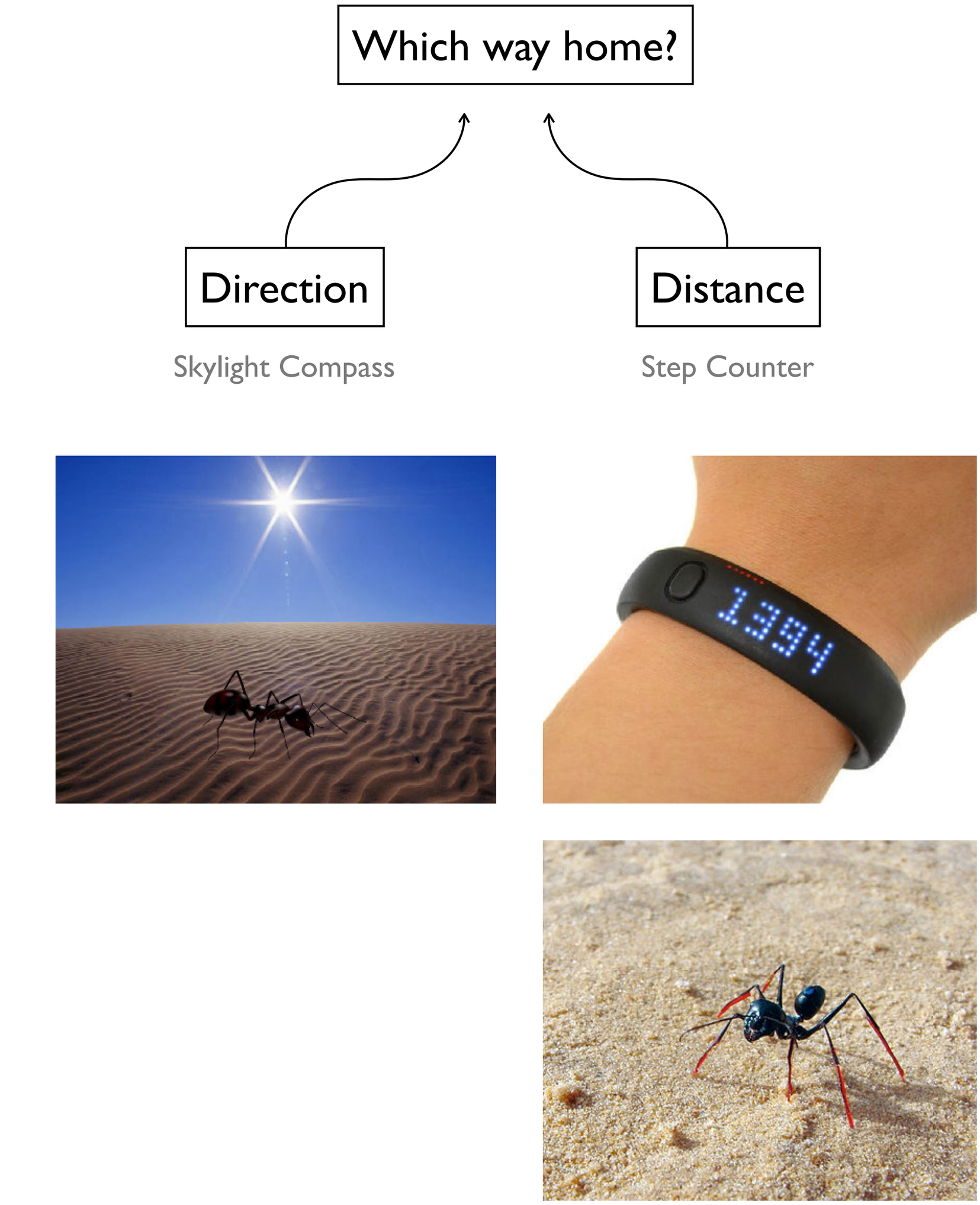
The Representational Hypothesis

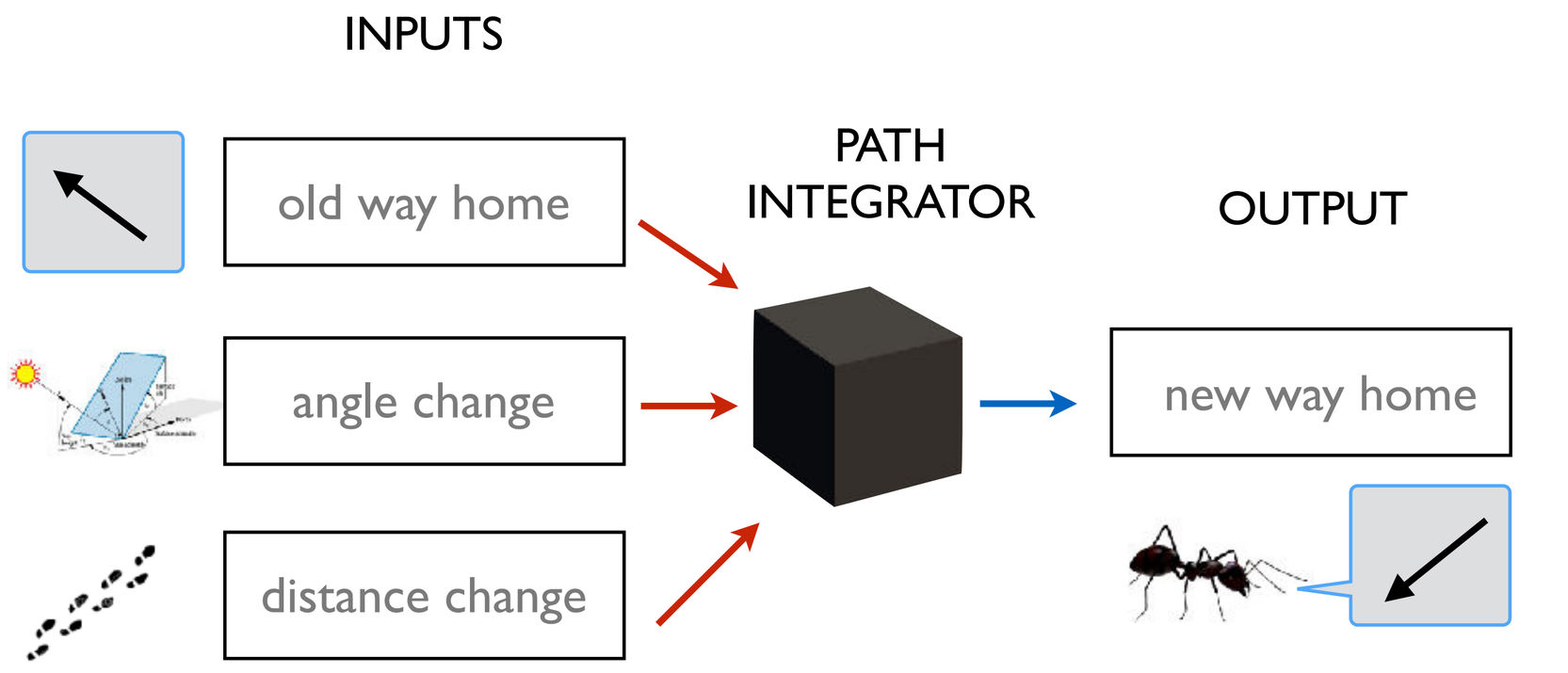


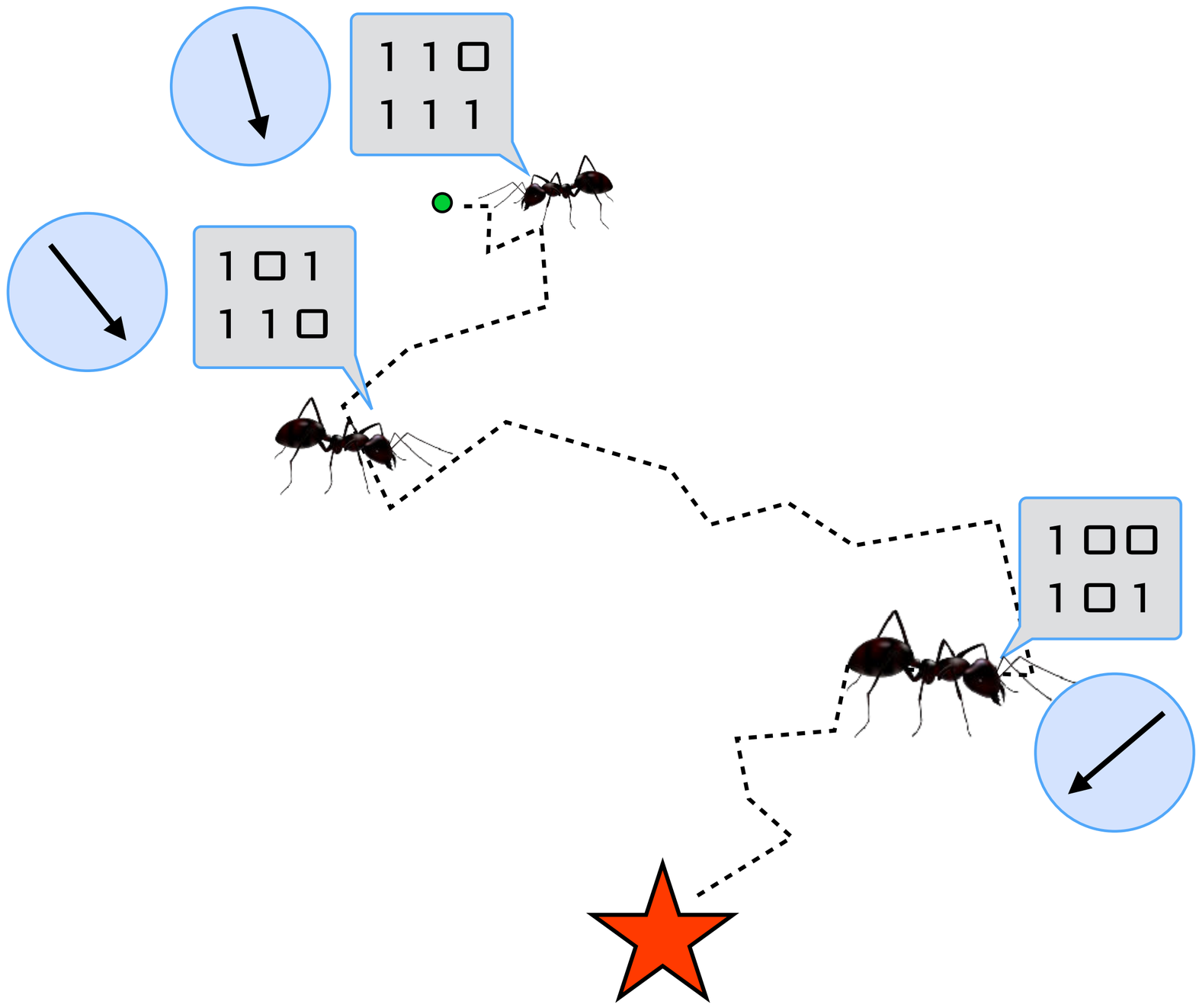
"**Dead reckoning**… is the method of navigating where no reference is made to landmarks. Instead, a record is kept of one's position in respect to some reference point by taking account of the distance that has been traveled and the changes in direction that have been made." Pearce 2008

Testing the Representational Hypothesis




Which way home?
Direction
Distance
Skylight Compass
Step Counter
13:34


INPUTS
old way home
angle change
distance change
PATH
INTEGRATOR
OUTPUT
new way home



The Desert Ant

First, search until you find food (are placed over it). Second, return to where you started as efficiently as possible.

